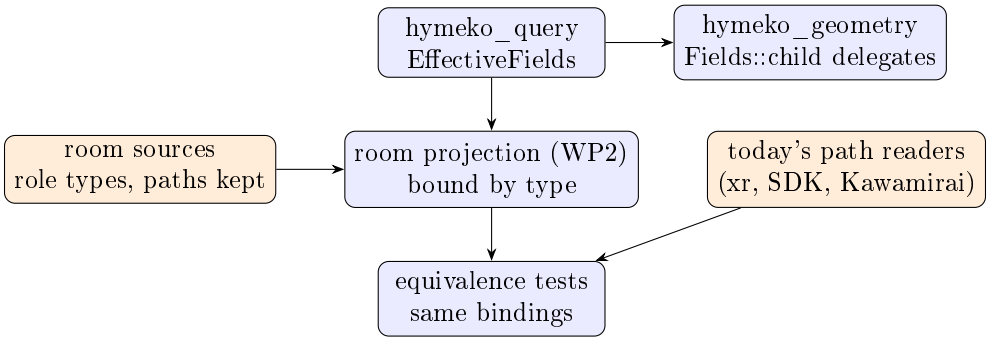
Declarative room structure (P1a)

Created-at: 2026-09-27 02:45 +02:00. Authorization 2026-09-27 (after 02:13): P1a now, inherited-field resolution in `hymeko_query`; (after 02:20): commit the uncommitted Transformation and SDK state first as separate prior commits (done: `transformation` dc002b2, `hymeko_web` da94b0b, both on `feat/declarative-room`). Standing rule: commit and push, new feat branch per repository. Predicates, declared conditions and HySTL stay with the RL line.

ETA WP1 1–1.5 h (until about 04:15). WP2–WP4 3–5 h after the vocabulary decision below. Basis: WP1 is one extracted reader, one delegation and focused tests; WP2 is a projection of the size of `SpatialQuery` plus two model sources and equivalence tests.

Findings that shape the plan

1. The own-then-bases field reader exists once, crate-private: `hymeko_geometry::kinematics::actuation::Fields` (own declaration first, then bases in authored order, cycle-safe). `hymeko_webxr`'s `attr_value`, `hymeko_query`'s `MatchContext::get_field` and the SDK inspector read direct children only; inheritance tests exist in three places (`decl_inherits`, `isa_transitive`, `check_inherits_simple`).
2. Two marker vocabularies coexist: ROS (`ros.marker`: `dictionary`, `marker_id`, `size_m`; `ros.registration`) used by the deployed room, and the spatial one (`spatial.kinds.fiducial_anchor`: `family`, `identifier`, `size_m`; `@fiducial_registration`) with a type-bound query (`hymeko_webxr::SpatialQuery`).
3. The ecosystem map lists `hymeko_webxr` as used by `hymeko_quest`; no XR, SDK, Kawamirai or HIVE source uses it (finding: map and code disagree).
4. `hymeko_query`'s integration target includes `bench_workflow`, an un-ignored timing `#[test]` (finding; not changed by this plan).



Work packages

WP1 `hymeko_query`: public effective-field reader (own declaration, then bases in authored order, cycle-safe, bounded), with a dotted-path form on `MatchContext` that is opt-in (the existing `get_field` keeps its direct-child semantics, so template emitters do not change output). `hymeko_geometry::Fields::child` delegates to it.

WP2 Room projection (*decision pending*): typed views of room root marker, landmarks, tracked headset, robot marker and editable room objects, bound by type through queries, never by path. Option A: extend `hymeko_webxr` (spatial owner per ecosystem map and AGENTS.md) and align the ROS marker with `fiducial_anchor`. Option B: room vocabulary beside the ROS one, projection in the SDK next to `RoomLayout`.

WP3 Model sources: role types on the deployed room (paths unchanged) and a typed rewrite of `hive-room.hymeko`; CLI gate for each file.

WP4 Equivalence tests: for both sources, today's path-based readers and the new projection yield the same bindings; README, symbols, duplication, report; commits and push per repository.

Affected files (WP1) `hymeko_query/src/{lib.rs, fields.rs (new), rewrite/match_context.rs}`, `hymeko_geometry/src/kinematics/actuation.rs`, tests in both crates, `SYMBOLS.md`. WP2–WP4 files are listed in the amendment after the decision. **CORE.YAML items touched:** none (Transformation declares none; Core is not edited).

Interface changes (WP1) New `hymeko_query::EffectiveFields` (child, value); `MatchContext::effective_fiel` No existing signature changes.

Amendment 2026-09-27 02:47 +02:00 A separate `EffectiveFields` struct repeated `QueryEngine`'s `ir`/`resolver` holder (token clone, +2/+16 against dc002b2). The reader is therefore `QueryEngine::effective_child( key)` and `effective_value(decl, key)` (module `fields.rs`); `QueryEngine::ir()` and `resolver()` now return the engine's `'a` borrow, so a value outlives a temporary engine (widening, source-compatible for callers). Duplication back to 384 clones / 6053 lines.

Amendment 2026-09-27 03:11 +02:00 (WP2–WP4, user decisions after 02:51) Option A: the room builds on the spatial vocabulary in `hymeko_webxr`. Approved dependencies: `hymeko_webxr` → `hymeko_query`; `hymeko_quest` → `hymeko_webxr`; generated, byte-checked vocabulary copies beside the room sources. ETA 3.5–4 h (until about 07:00).

- **WP2** (`transformation`): new `data/webxr/meta_room.hymeko` importing `meta_spatial`: kinds `room` (scene), `room_frame` (spatial frame), `aruco_marker` (fiducial anchor with defaults `fiducial_kind "aruco"`, `convention`), `tracked_device` (spatial frame). No observation slot on a kind: a slot inherited from a kind would be shared by every instance. The root is expressed as a relation, reusing `@fiducial_registration (+ room_frame, ~ marker)`. New `RoomQuery` (`src/room.rs`): rooms, the room frame (exactly one), the optional root registration (at most one), landmarks (other fiducials inside the room), tracked devices, and all ArUco markers of the model, each with its own `sample` slot; fields read through `QueryEngine::effective_value`. `SpatialQuery`'s attribute reads use the same effective rule (a kind's default is inherited; the validator's reserved-field check stays direct). Vocabulary texts exported as constants for the copy check.

- **WP3** (`middleware_hymeko`, new `feat/declarative-room`): copies of `meta_spatial.hymeko` and `meta_room.hymeko` in `examples/`, written by `hymeko_webxr/tools/sync-vocabulary.ps1`; `ros-ur5.hymeko` imports them; the room becomes `+ <isa> room` with a room frame and the root registration; the five markers and the headset gain the spatial bases; markers gain `family`/`identifier` beside `dictionary`/`marker_id` (expand step; readers switch in P2). Paths unchanged; `ros.meta.hymeko` unchanged (two middleware tests compile it standalone and imports are not transitive). `hive-room.hymeko` unchanged: it declares no marker or device. Room objects stay with `RoomLayout` (the kit is appended to arbitrary workspaces and cannot import the vocabulary); typing them is a P2 item.

- **WP4** (`xr`, new `feat/declarative-room`): `hymeko_quest` test dependency use of `RoomQuery`: for the UR5 room and `hive-room`, the path readers in `lan.rs` (`marker_at`, landmarks, observer slot) and `RoomQuery` yield the same root, landmark set and headset slot; `family`/`identifier` equal `dictionary`/`marker_id`; the middleware copies equal the exported vocabulary. CLI gate on `meta_room.hymeko` and `ros-ur5.hymeko`; middleware tests re-run.

Affected files: `transformation/hymeko_webxr/{Cargo.toml, src/lib.rs, src/query.rs, src/room.rs, data/webxr/meta_room.hymeko, tools/sync-vocabulary.ps1, tests}`, `Cargo.lock`; `middleware_hymeko/example meta_spatial.hymeko, meta_room.hymeko}`, README; `xr/hymeko_quest/{Cargo.toml, src/lan.rs}`, `Cargo.lock`, README. CORE.YAML items touched: none.

Tests Unit: own value overrides a base; bases in authored order; multi-level inheritance; diamond and cycle terminate; missing field; dotted path through an inherited block. Regression: `hymeko_geometry` inherited-material/geometry and actuation tests unchanged. Correctness only; `--skip bench_` for the query integration target; no measurement.

Budgets Reader cost is bounded by the base graph (visited set); no allocation per direct hit beyond the walk stack. No performance measurement is authorized.

Rollback Revert the WP1 commit; geometry returns to its private reader.

Risks (1) Changing `get_field` globally would change emitted URDF/SDF; the new reader is opt-in. (2) The vocabulary decision changes a deployed model (WP3); live HIVE room migration is P2 and not part of this plan. (3) Dirty sibling state is now committed but still depends on uncommitted Core APIs; builds are local integration builds.